

one UrbanGraph → many consumers, no conversion

1 · Edit table

ordinary GeoPandas on *edges_gdf*

type	route	t_min
walk	-	0.8
boarding	1	1.0 → 6.0
tram	1	2.4
boarding	1	1.0 → 6.0

```
mask = (type=="boarding")  
      & (route=="1")  
edges.loc[mask,"time_min"] += 5
```

≤ 0.08 s

2 · Rebuild

`g.update_adjacency_matrix()`

cached matrix
rebuilt from
the edge table

≈ 1.3 s

3 · UrbanGraph

same object

- node & edge tables
- cached matrix

**no format
conversion**

≈ 5–7 s

IduEdu routing

- nearest-source, OD
- re-screened
accessibility maps

ObjectNat

- `get_graph_isochrones`
- `get_graph_coverage`
- polygon products

■ provided by IduEdu ■ downstream (ObjectNat) ■ ordinary GeoPandas / user code